

The Impact of Weather Factors on Tourism in Brisbane

Group Chimera

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Abstract

As tourism plays a crucial role in Brisbane's economy system, it is important to explore the reasons behind the changes in the number of tourists visiting the city. This study focused on exploring the impact of weather conditions on tourism, examining whether rain, temperature, and seasons contribute to the changes in tourist number using three models. The first model considered rainfall as the main factor, then temperature factor was added in the second model, and subsequently the monthly seasonal factor was taken into consideration in the third model. These models were created using two main approaches which are correlation and multiple regression.

The findings from this study suggested that rainfall and temperature had minimal effect on the number of tourists, while the seasonal factor considerably contributed to the change in tourist number. This indicates that travelling decisions are not only influenced by a single factor such as rainfall amount, but by multiple weather conditions. The study contributes to supporting tourism businesses and government in improving tourism management.

Keywords: Tourism demand, visitor numbers, rainfall, temperature, seasonality, multiple linear regression

GitHub Repository

The GitHub repository for this project contains the datasets, data cleaning process, analysis scripts, visualizations, and report files used in this study.

Repository Link: <https://github.com/melody911020-bot/Chimera-Group>

Introduction

Tourism demand can be affected by many factors. These factors include weather conditions and seasonal travel patterns. For Brisbane, the weather is very important because the city has many outdoor tourism activities and these activities include riverside recreation, outdoor dining, nearby islands, national parks, and adventure activities (Tourism Australia, n.d.). Therefore, rainfall and temperature are relevant weather variables in this study. Although this project focuses on tourism, previous research in Brisbane has shown that weather can influence people's travel plans (Wei, 2022).

Warm weather may make the vacation more enjoyable, because it is usually more suitable for outdoor activities. Heavy rainfall has the opposite effect; it may reduce tourists' willingness to visit some outdoor places. However, weather does not always affect tourism directly. Rainy or hot periods may also happen during busy holiday seasons. Therefore, it is important to consider seasonality when analyzing visitors' demand.

Previous studies have shown that weather and climate can affect tourism demand. For example, Liu (2016) found that rainfall and temperature influenced visitor numbers in Taiwan national parks. Seasonality is also an important factor in tourism because visitor numbers often change across different months and travel seasons (Senbeto et al., 2024). Weather may influence tourists' travel choices and School breaks, public holidays, and busy travel seasons can also lead to changes in visitor numbers. That means rainfall or temperature alone may not fully explain tourism demand. And visitor numbers may change because of monthly travel patterns, not only

because of weather conditions. Therefore, adding seasonal factors can reduce deviation of the analysis.

The aim of this project is to investigate how rainfall, temperature, and seasonality affected tourist numbers in Brisbane. This study uses statistical analysis to see if rainfall alone affects tourist numbers. It also tests model performance with added temperature and seasonal factors through regression comparison. This project aims to identify if weather variables or seasonal factors are more useful to explain tourism demand. The results can help local tourism operators and government have better predictions of visitor numbers and improve the development of tourism.

Data

This study used monthly climate and tourism data from January 1996 to December 2025 in Brisbane, Queensland, Australia. Rainfall data were collected from the Alderley weather station through the Australian Bureau of Meteorology (BOM). Due to missing observations, rainfall data from February 2017 to December 2017 were supplemented using data from the Brisbane Botanic Gardens Mt Coot-tha station. Monthly temperature data was obtained from the Archerfield Airport weather station, also provided by the BOM. Monthly inbound passenger data for Brisbane Airport was extracted from the BITRE airport traffic statistics database.

All datasets were collected from official public sources and merged into a single monthly dataset for analysis. Data cleaning procedures included checking for missing values, standardizing date formats, converting variables into numeric form, and aligning all datasets by year and month before analysis. The study area focuses on Brisbane, the capital city of Queensland, which is one of Australia's major tourism and transportation hubs.

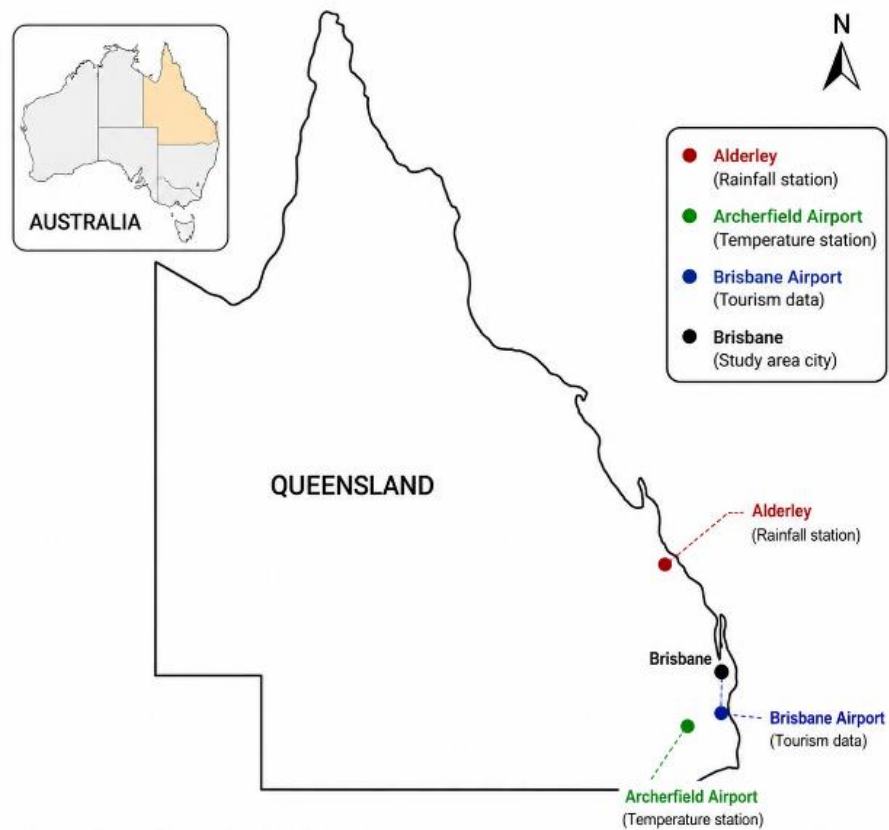


Figure 1 Study area and locations of climate observation stations and Brisbane Airport in Queensland, Australia.

Source: Adapted from WorldAtlas.com with climate and tourism data locations based on BOM and BITRE datasets.

Methods

This analysis aims to examine the combined effects of meteorological factors—such as precipitation and temperature—and seasonal cycles on fluctuations in visitor numbers. By employing a stepwise multiple linear regression model and incorporating control variables, we aim to eliminate the effects of multicollinearity and seasonal interference, thereby quantifying the impact of each individual factor on tourism demand.

This report employs the following statistical methods:

- **Correlation Analysis:** Uses correlation matrix to assess bivariate linear relationships between continuous variables.
- **Hierarchical Multiple Linear Regression:** modelling strategy that progressively introduces additional control variables to assess the model's explanatory power.
- **Dummy variable regression:** Convert the time periods into dummy variables and include them in the model to explain and control for fluctuations caused by these factors.

Analytical workflow:

1. Exploratory Analysis:

The data is analysed using the Pearson correlation coefficient to provide a comprehensive overview. This helps identify potential risks of multicollinearity among the independent variables in advance, thereby ensuring the robustness of the subsequent multiple regression model.

2. Hierarchical Modelling:

First, Baseline model containing only single meteorological factor was established.

Subsequently, it was upgraded to a comprehensive climate model that integrated multiple meteorological indicators, and finally, a seasonal fixed-effects model was introduced.

Analysis of variance ANOVA was used to compare the goodness of fit of the models at different levels.

3. Incorporating seasonal effects:

Given the tourism industry's high dependence on seasonal fluctuations, monthly characteristics are transformed into independent variables and introduced into the model as control variables.

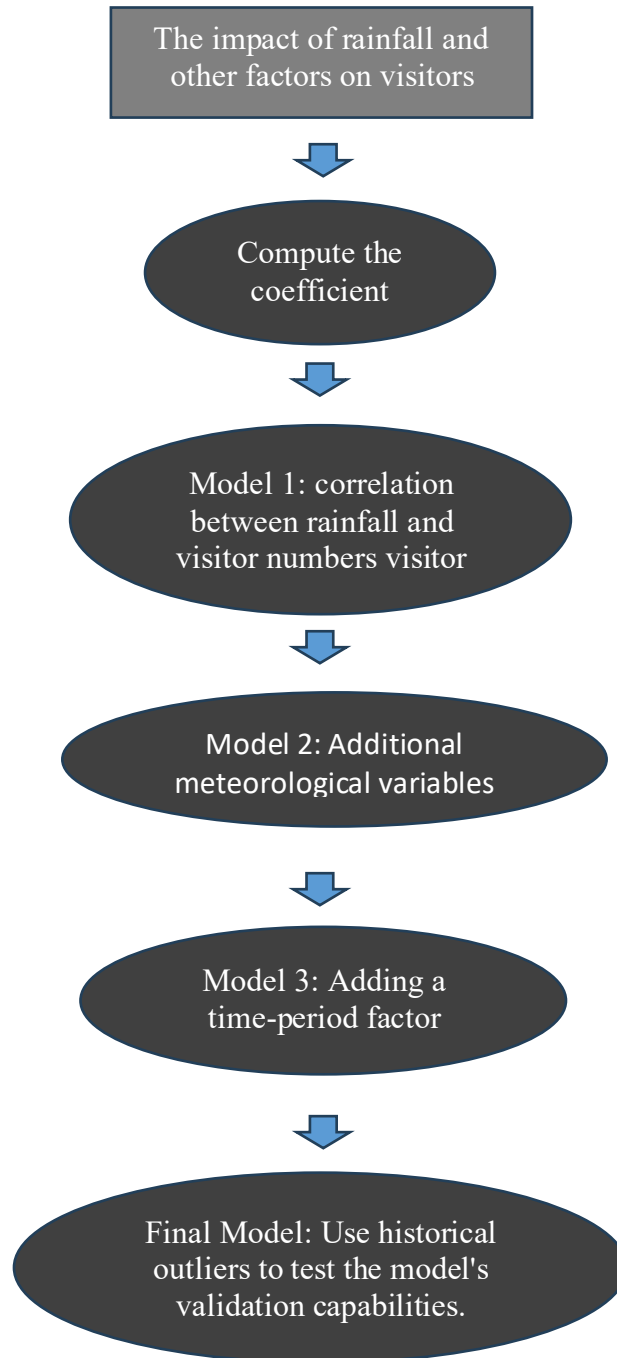


Figure 2 Flow Chart: The impact of rainfall and other factors on visitors

Results

Correlation:

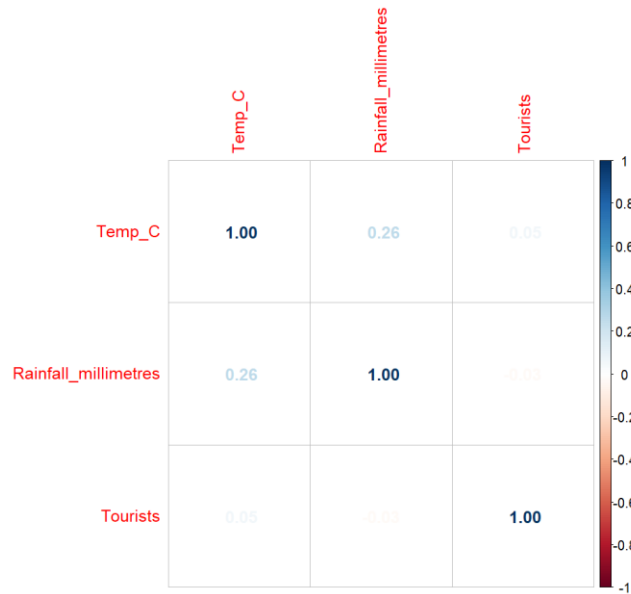


Figure 3 correlation matrix

According to figure 3 we can find that rainfall is negatively related to tourists, while temperature shows a positive relationship with tourism demand. At the same time, there is a weak positive correlation between temperatures and rainfall. Therefore, we need to consider the impact of temperatures and maybe there are some complex factors that exist in them, because of their weak relationship.

Model 1: Rainfall Only

The simple regression result shows that rainfall has a negative but statistically insignificant relationship with tourist numbers. The coefficient indicates that a 1 mm increase in rainfall is associated with an average decrease of about 62 tourists (rainfall coefficient = -62.06).

However, the p-value is greater than 0.05 (p-value = 0.587), suggesting that rainfall alone does

not significantly explain changes in tourist numbers. The very low R-squared value ($R^2 = 0.0008$) also indicates that rainfall explains only a negligible proportion of tourism demand variation.

Model 2: Rainfall and Temperature

The regression results show that neither rainfall nor temperature has a statistically significant effect on tourist numbers. Although rainfall has a negative coefficient (estimate = -97.08 , p-value = 0.412), its effect is not significant. Temperature showed a positive coefficient (estimate = $4,527.87$, p-value = 0.252) but was also non-significant. The very low R-squared value ($R^2 = 0.004$) suggests that rainfall and temperature alone explain only a small proportion of the variation in tourist numbers. Therefore, additional factors such as seasonality, holidays, economic conditions, and travel restrictions may need to be included.

Model 3: Seasonal Controls

To study the seasonal effects, Model 3 introduced a month factor, along with rainfall and temperature, to predict changes in tourist numbers. The R-squared value increased to 0.0973 , indicating that the combination of weather indicators and seasonal controls explains approximately 9.73% of the total variance in tourist numbers.

Effect of Temperature. Controlling for seasonality and rainfall, temperature has a significant positive impact; each 1°C increase caused $56,351$ tourists increasing.

Effect of Rainfall. After controlling temperature and month, the coefficient for rainfall changed

from negative to positive, but it was still not significant ($p = 0.137 > 0.05$). Therefore, it cannot be proven that rainfall would deter tourists from visiting.

Seasonal Variations. The monthly variables exhibited a clear seasonal model.

Table 1

Significant Monthly Coefficients Relative to January (Model 3)

Month	Coefficient	Explanation
June	+476,328	Much more than in January
July	+590,337	Highest, significant
August	+435,422	Significant
September	+306,412	Significant
October	+244,579	Significant

After controlling temperature and monthly seasonality, the overall regression model becomes statistically significant. Rainfall remains statistically insignificant, suggesting that rainfall alone may not be a strong predictor of tourist numbers. In contrast, temperature shows a significant positive relationship with tourist numbers. The inclusion of monthly dummy variables significantly improves the model, indicating that seasonality plays an important role in tourism demand.

Conclusion

This study examined the relationship between weather and tourism demand in Brisbane by correlation analysis and multiple regression models.

The result showed that, when we considered rainfall alone, it didn't explain much variation in tourist numbers. Rainfall showed a weak negative relationship with tourism demand regardless of whether temperature or seasonal factors were included.

In model 2, temperature showed a weak positive relationship with tourist numbers, but became more pronounced after the seasonal factor was added to the model.

After adding monthly dummy variables, the regression model improved, which suggested that the seasonal factor plays an important role in explaining tourism demand. This also showed that tourist numbers showed a clear upward trend from June to October.

There are several limitations in our study. First, weather variables overlap with the seasonal factor. Second, this study didn't consider other factors, such as pandemics, exchange rates, natural disasters, or wars; these factors might also affect the tourism demand. In addition, the tourism data may not fully capture tourists who use other transport methods.

Future research could include other conditions and different methods of transportation to improve the explanatory power of the models. More advanced methods, such as adding machine learning for forecasting tourism demand.

Through this research, local tourism businesses can improve human resource arrangements,

accommodation capacity management, and marketing strategies. The government may also improve tourism demand forecasting to support transportation management, infrastructure planning, and tourism management.

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